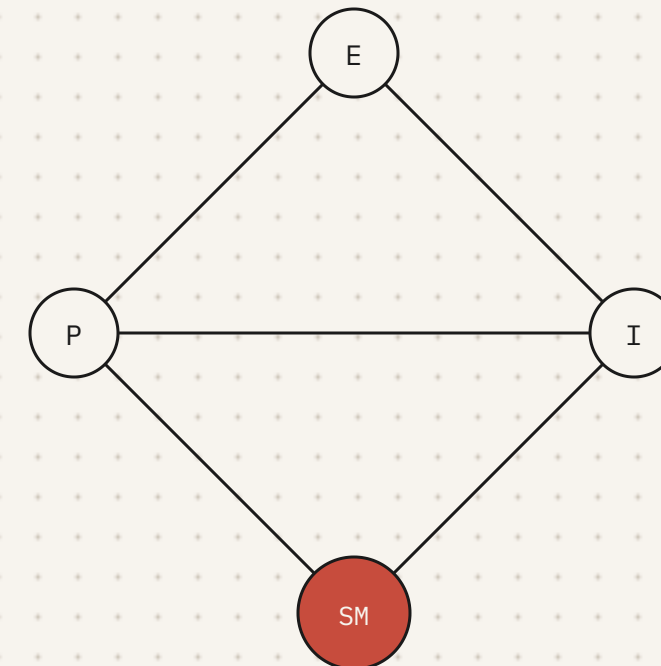


A CONTROLLED STUDY OF INTER-AGENT COMMUNICATION

Communication protocol effects on multi-agent system efficiency and task completion.

How four message formats — natural language, Markdown, JSON, and a shared-memory blackboard — change cost and quality in a fixed three-agent LLM pipeline across math, reading, and news.



AUTHORS

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DATE

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DEMO ARTIFACT

FASTAPI DASHBOARD · FASTAPI_APP.PY

MOTIVATION

Frameworks fix what agents *do*. They leave open how those agents *talk*.

LangGraph, AutoGen and CrewAI prescribe roles, tools and orchestration. The wire format between agents is a free parameter — and one practitioners pick by habit.

We treat protocol as a controllable factor and ask whether it materially changes cost and quality.

OUR CONTRIBUTIONS

- ▶ **Controlled protocol-level benchmark** — four formats × three domains, model and prompts held fixed.
- ▶ **360-run dataset** with full message-level logs, persisted per agent call.
- ▶ **Two-way ANOVA, Tukey HSD and bootstrap 95% CIs** — uncommon in LLM systems papers.
- ▶ **FastAPI dashboard** — the recommended demo artifact, with live protocol recommendations.

No prior published study compares these wire formats head-to-head with a controlled experimental design.

RESEARCH QUESTIONS

Two questions, one factorial design.

RQ1_{EFFICIENCY}

Does communication protocol affect **token cost and latency** in a multi-agent pipeline, holding roles, prompts and model fixed?

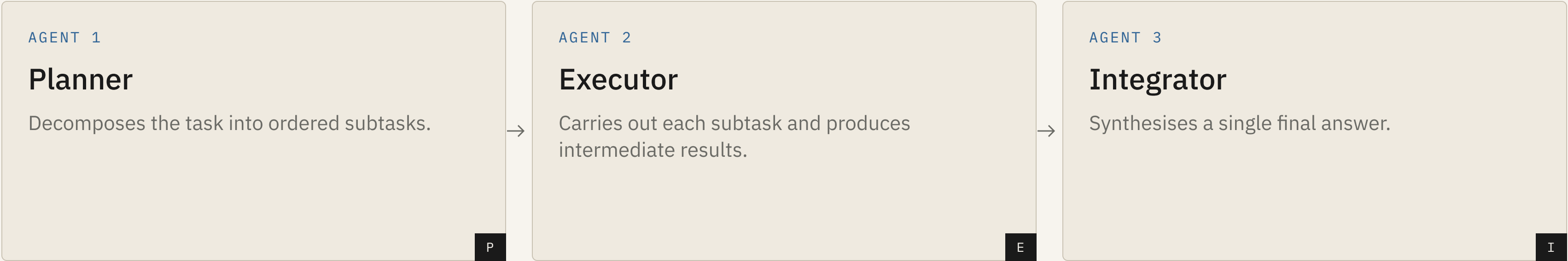
RQ2_{EFFECTIVENESS · INTERACTION}

Does protocol affect **task completion**, and does the best protocol **depend on the task domain**?

Answered with a 4×3 factorial — four protocols, three task domains — and a two-way ANOVA on tokens, latency and completion score.

EXPERIMENTAL SETUP

A fixed three-agent pipeline; only the protocol varies.



HELD CONSTANT

Roles, system prompts, max-tokens caps, temperature 0.3, model gpt-4o-mini, OpenAI seed=rep.

MANIPULATED

Inter-agent protocol — NL ·
Markdown · JSON ·
Shared Memory.

DESIGN

$4 \times 3 \times 10$ samples $\times$ 3 reps = **360 runs**, 1 080 API calls.

THE FOUR PROTOCOLS

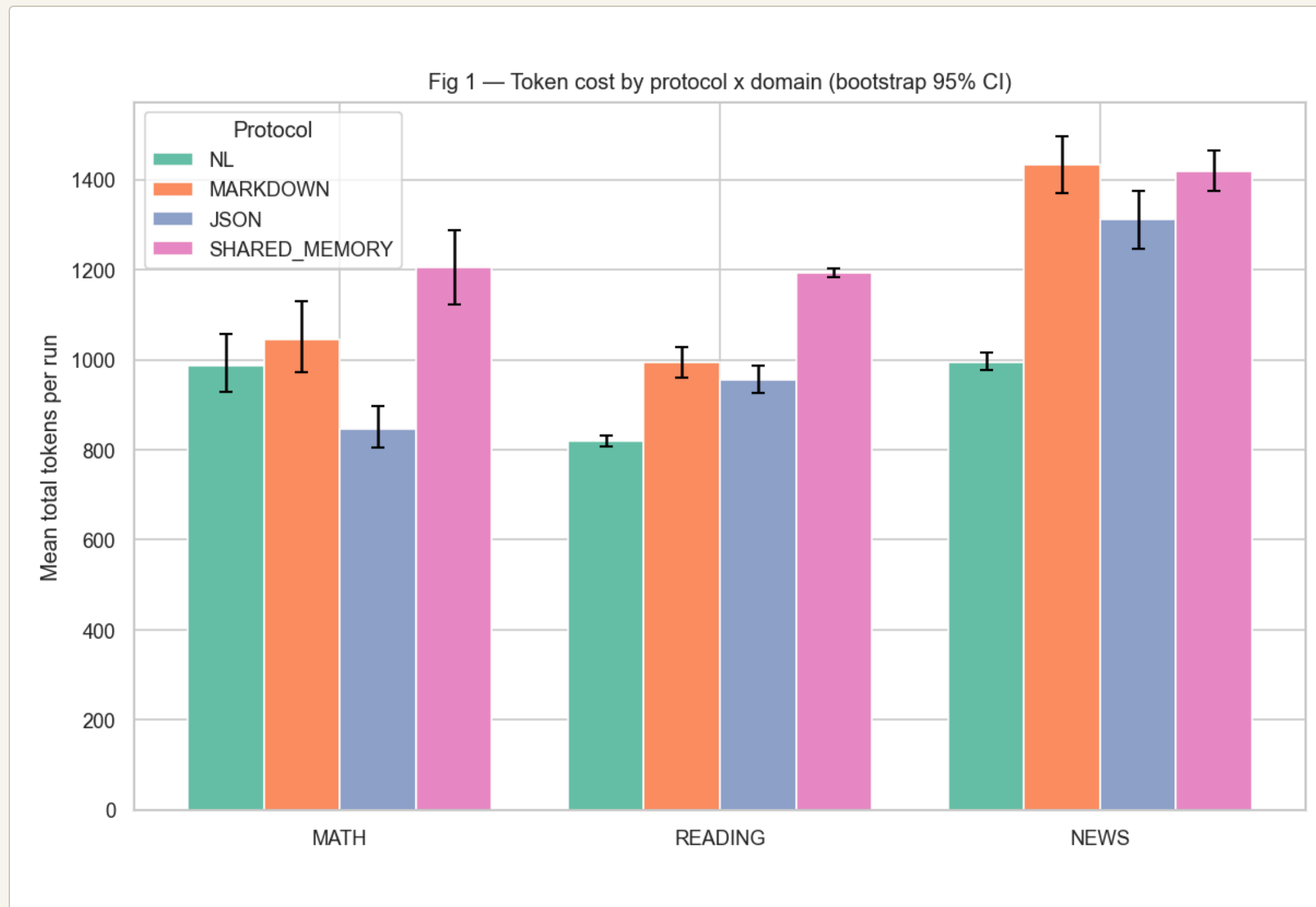
Same pipeline. Four different wire formats.

● NL	● MARKDOWN	● JSON	● SHARED_MEMORY
<pre>The planner first identified the question, then asked the executor to retrieve the relevant span from the passage and return it as a short phrase.</pre>	<pre>## Plan 1. Identify question 2. Locate span in passage 3. Return as phrase ## Result - Denver Broncos</pre>	<pre>{ "plan": ["identify q", "locate span"], "answer": "Denver Broncos" }</pre>	<pre>{ "plan": {...}, "execution": {...}, "context": "<full passage>" } // re-injected into every agent</pre>
Plain prose. Implicit default. Cheapest input framing.	Headings + bullets. Inflates output tokens.	Schema-enforced. response_format. 0 / 90 unrecoverable parse errors.	Blackboard. Full state to all agents. Largest input tokens by 2.4×.

Mean integrator prompt tokens – NL 194 · Markdown 254 · JSON 188 · Shared Memory 469

RQ1 • EFFICIENCY

Protocol drives a quarter of token-cost variance.



TWO-WAY ANOVA • TOTAL TOKENS

$$\eta^2_{\text{protocol}} = 0.262$$

$$F = 87.4 \cdot p < .001$$

TUKEY HSD • POOLED

All six protocol pairs differ. **Shared Memory costs +338 tokens / run vs NL.**

POOLED ORDER

NL < JSON < MARKDOWN < SHARED_MEMORY

WHERE THE TOKENS GO

Shared memory pays in input. Markdown pays in output.

INPUT — PROMPT TOKENS

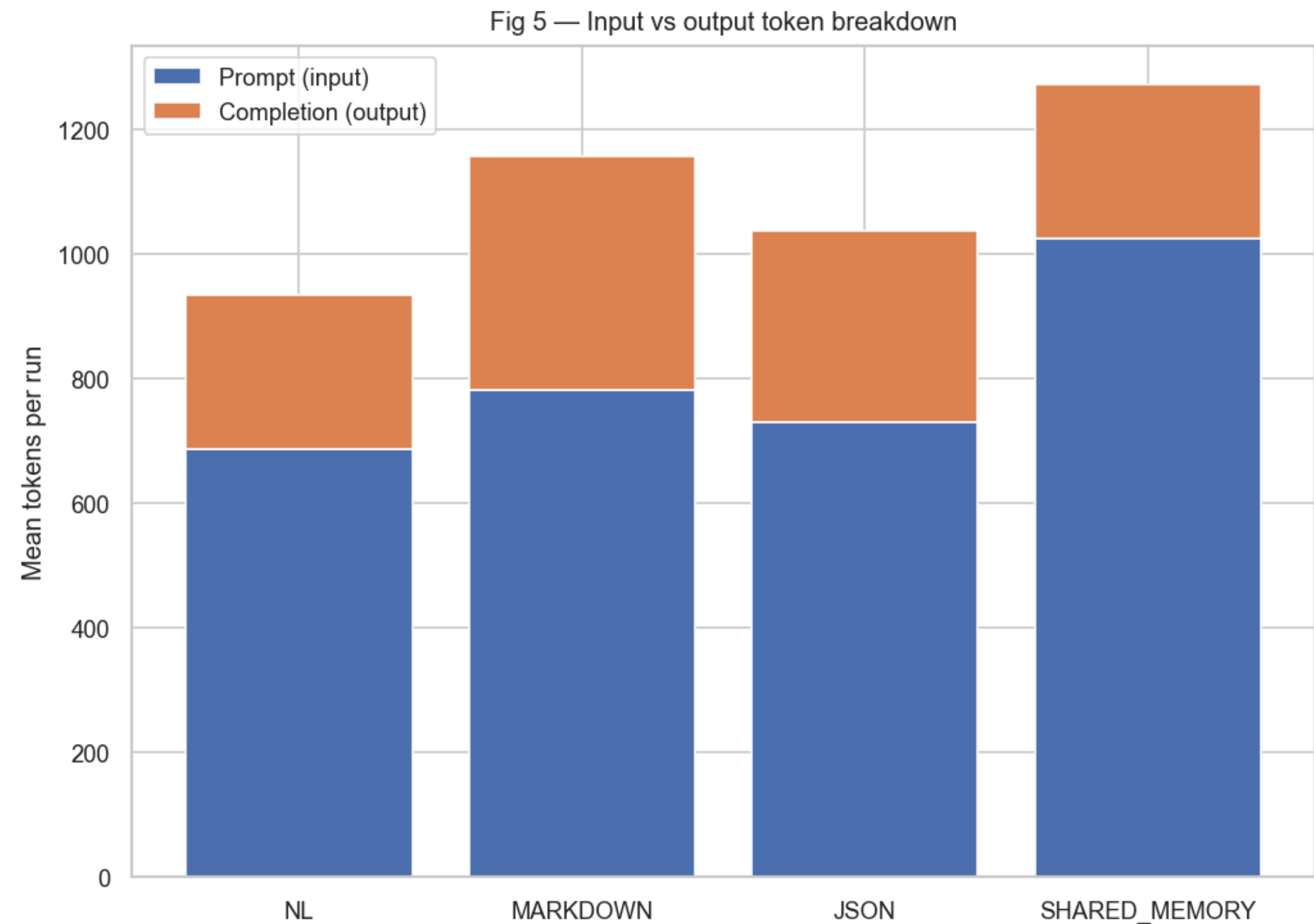
Shared Memory re-injects accumulated state on every agent call. **+50% input** over NL.

OUTPUT — COMPLETION TOKENS

Markdown's headings and bullets balloon executor output. JSON keeps it terse.

IMPLICATION

Schema tightness $\neq$ verbosity. Two distinct levers, two distinct protocols.

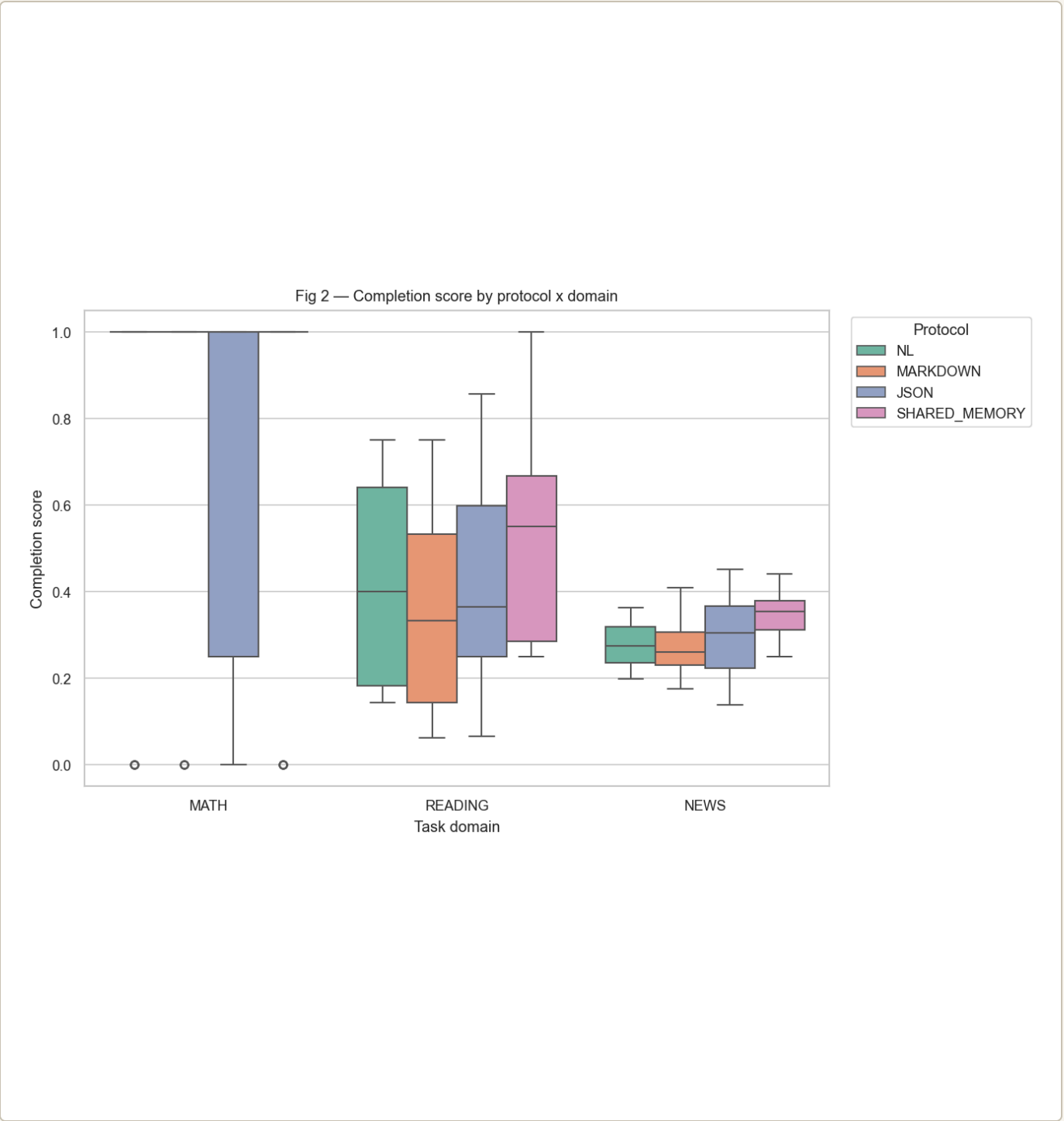


RQ2 • EFFECTIVENESS

Protocol matters for reading and news. It doesn't move math.

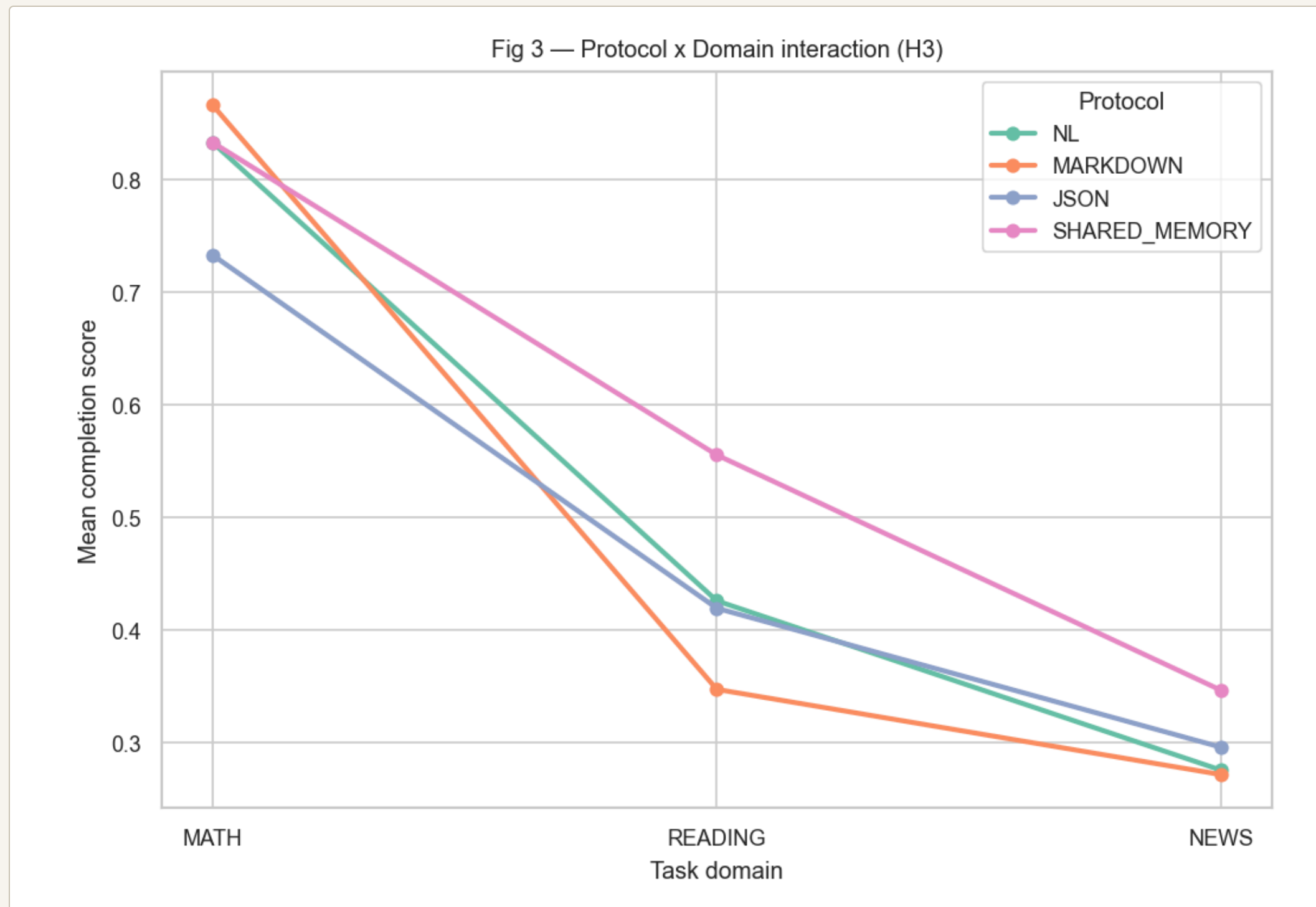
DOMAIN	F	P	H ²	COHEN D (best vs worst)	BEST
MATH	0.66	.58	.017	0.33	● MARKDOWN 0.867
READING	4.39	.006	.102	0.88	● SHARED_MEMORY 0.555
NEWS	9.51	<.001	.197	1.44	● SHARED_MEMORY 0.346

Per-domain one-way ANOVA on completion score · Tukey HSD confirms SHARED_MEMORY > MARKDOWN on both READING ($\Delta=+0.21$, $p=.003$) and NEWS ($\Delta=+0.075$, $p<.001$).



INTERACTION

The best protocol changes with the domain.



INTERACTION · TOKENS

$$F = 15.66 \cdot p < .001$$

$$\eta^2 = 0.094$$

PER-DOMAIN EFFECT-SIZE SWING

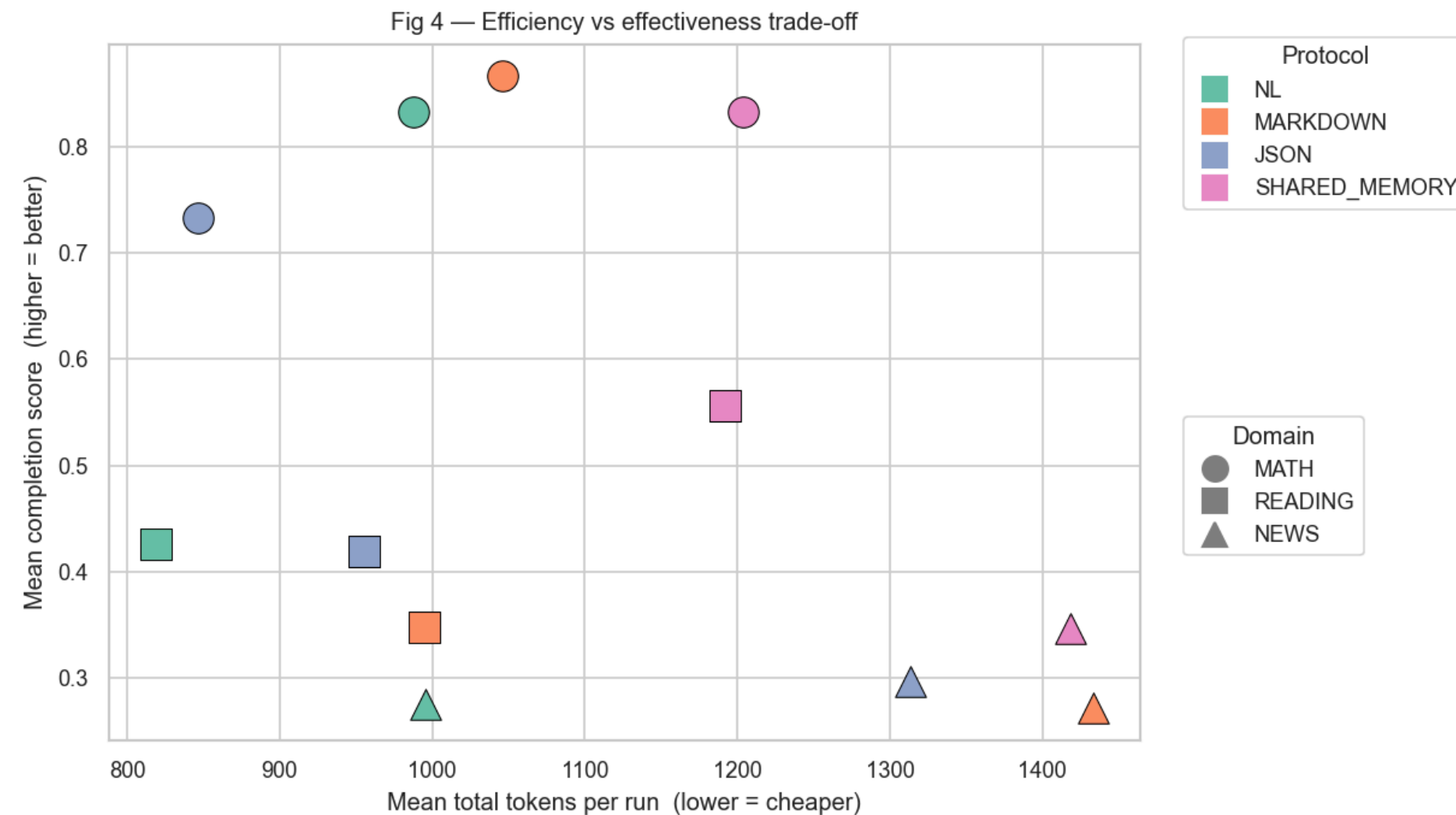
η^2 0.017 → 0.197 — a 10× change across domains.

READING THE LINES

Markdown wins MATH by ϵ . Shared Memory wins READING and NEWS by wide margins. Lines cross — no single protocol dominates.

EFFICIENCY VS EFFECTIVENESS

There is no universally best protocol — only a frontier.



PARETO-OPTIMAL CELLS

MATH · JSON (cheapest) → Markdown (best)

READING · NL (cheapest) → SM (best)

NEWS · NL (cheapest) → SM (best)

DOMINATED

Shared Memory on MATH (NL beats it on both axes).

Markdown on READING and NEWS.

SUPPLEMENTAL EXPERIMENTS

We isolated the mechanism — and validated across providers.

Full 2×4 mechanism × format ablation

+360 runs · gpt-4o-mini. Held output format fixed (NL / Markdown / JSON / Default), varied only the mechanism (relay vs blackboard).

Mechanism alone adds +512 to +786 tokens per run — Cohen $d > 3$, $p < .001$ across all 24 cells. No directional reversals.

Cleanly attributes H1 to state serialization, independent of output format.

DeepSeek V4 Flash robustness

+240 runs · 8 protocols × 3 domains × 5 samples × 2 reps on a different model provider.

Direction transfers — 24 / 24 cells preserved sign. Magnitudes within ~20% of OpenAI on most cells.

Confirms external validity beyond gpt-4o-mini.

PRACTITIONER RECOMMENDATION

Pick the protocol per domain, not per pipeline.

• MATH

JSON or Markdown

JSON is cheapest at **847 tokens**; Markdown gives the highest completion (0.867) for +24% tokens. All four protocols are statistically tied on math accuracy.

Skip Shared Memory — dominated.

• READING

SHARED MEMORY

+30% completion over Markdown for +45% tokens (Cohen d = 0.88). The blackboard preserves exact spans the integrator quotes verbatim.

Budget alt: NL — cheapest *and* second-best completion.

• NEWS

SHARED MEMORY

The only protocol that materially lifts completion on summarisation (Cohen d = 1.44 vs Markdown). Worth the +25% input tokens.

Budget alt: NL — strong default at lowest cost.

LIMITATIONS

What this study does *not* answer.

Scale

N = 10 samples × 3 reps per cell. Pooled Protocol × Domain interaction on completion is underpowered (p = .203) despite per-domain effect sizes ranging 0.017 → 0.197.

Protocol coverage

Full 2×4 mechanism × format ablation completed. XML, YAML, hybrid formats, mid-pipeline switching, and learned protocols remain untested.

Cross-provider scope

Validated across DeepSeek V4 Flash (24 / 24 cells preserved sign, magnitude drift ≤ 20% on most). Anthropic, Gemini, and open-source model families remain untested.

Truncation tail on Markdown

11/360 runs hit max_tokens; 6 of those are Markdown · MATH (20%). Output ceiling co-varies with format choice.

CONCLUSION

Protocol is a measurable design choice — and it should be picked per domain.

H1 · EFFICIENCY

Supported.

Protocol explains 26% of token-cost variance. Shared Memory is most expensive on every domain; JSON is cheapest on math.

H2 · EFFECTIVENESS

Direction reversed.

Protocol matters most for news ($\eta^2=.20$), then reading (.10), then math (.02). The proposal predicted the opposite.

H3 · INTERACTION

Domain-dependent.

Best protocol changes with the task. Shared Memory wins reading and news; JSON or Markdown wins math.

TAKEAWAY *Protocol is not formatting polish — it is an experimental design choice.*